CARBIMAZOLE

Agranulocytosis — sore throat

Avoid in 1st trimester

CHEMICAL NAME Ethyl 3-methyl-2-sulfanylidene-2,3-dihydro-1H-imidazole-1-carboxylate

CLASS Thionamide antithyroid drug (prodrug of thiamazole)

TRADE NAMES Neo-Mercazole · Carbimazole (generic) · Neomercazole

THERAPEUTIC USES / INDICATIONS

- Hyperthyroidism (Graves' disease, toxic nodular goitre).
- Preparation for thyroidectomy or radioiodine.
- Block-and-replace regimen (with levothyroxine).

ADMINISTRATION

- **PO** once daily or in divided doses.
- Titration regimen or block-and-replace.
- Treatment usually 12–18 months for Graves'.

DOSAGES

- **Initial:** 15–40 mg daily (higher in severe cases), until euthyroid (4–8 weeks).
- **Maintenance (titration):** 5–15 mg daily.
- **Block-and-replace:** 40–60 mg daily + levothyroxine 50–150 mcg.
- **Children:** 250 mcg/kg three times daily initially.

PHARMACOKINETICS

- **Onset:** 1–2 weeks (pre-formed hormone stores); euthyroid 4–8 weeks.
- Converted rapidly to thiamazole (methimazole); concentrated in thyroid.
- Excreted in urine. **t½:** 6 h (intrathyroidal action longer).

PHARMACODYNAMICS

- Inhibits **thyroid peroxidase** → blocks iodination of tyrosine and coupling → ↓ T4 and T3 synthesis.
- Does not affect release of stored hormone.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, mild GI upset, headache.
- Rash, pruritus.
- Arthralgia; taste disturbance.

ADVERSE EFFECTS (SERIOUS)

- Agranulocytosis/neutropenia (usually first 3 months) — sore throat, fever, mouth ulcers.
- Hepatotoxicity (cholestatic).
- Acute pancreatitis.
- Teratogenic (aplasia cutis, embryopathy) in first trimester.

CONTRAINDICATIONS & CAUTIONS

- Severe blood disorders; previous agranulocytosis with carbimazole.
- Severe hepatic impairment.
- First trimester of pregnancy (use propylthiouracil).
- History of pancreatitis with carbimazole.

KEY INTERACTIONS

- **Warfarin:** effect may change as thyroid status normalises.
- **Digoxin, theophylline:** dose needs may fall as patient becomes euthyroid.

NURSING MANAGEMENT

- Warn every patient about sore throat/fever — stop drug and check urgent FBC.
- Monitor TFTs every 4–6 weeks during titration.
- Monitor LFTs if symptoms of liver disease.
- Monitor HR (β-blocker often co-prescribed early).
- Check pregnancy status.

PATIENT EDUCATION

- Stop and get an urgent blood count if you develop sore throat, fever, mouth ulcers or flu-like illness.
- Report yellow skin, dark urine, itching.
- Use contraception; tell your doctor if planning pregnancy.
- Treatment usually lasts 12–18 months.

KEY POINTS

- Thionamide; blocks thyroid peroxidase.
- Prodrug of thiamazole.
- Agranulocytosis: sore throat = urgent FBC.
- Avoid in 1st trimester (use PTU).